

Marketing Analytics

Chapter 3 · Uncertainty

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September 2026

Before we start

Paper quiz on Chapter 2 (*10 min*)

- Individual, closed book
- Counts toward participation

Assignment 1 is running

- Deadline: Group A Mon, Oct 26 · Group B Wed, Oct 28
- Debrief at the start of next session
- Questions? → forum on Canvas

Today's session

Part A — Uncertainty (*this deck*)

- 3.1 What a poll can and cannot tell you
- 3.2 Populations and samples
- 3.3 Sampling variation
- 3.4 The standard error
- 3.5 The central limit theorem
- 3.6 Confidence intervals
- 3.7 Shares and the margin of error
- 3.8 Uncertainty in R

Part B — Experiments I (*next deck*)

Comparing two groups: is the difference larger than sampling variation alone would produce?

Driving question: *How sure can I be?* Every number from a sample is an estimate — today we learn to say **how good** an estimate it is.

3.1 What a poll can and cannot tell you

“Plus or minus 3.1 percentage points”

“The poll was conducted Sept. 13–17 and surveyed 1,000 registered voters nationally. The margin of error is plus or minus 3.1 percentage points.”

— NBC News, September 2024

More than 150 million people voted. How can **1,000** respondents tell us anything? Where does ± 3.1 come from — and why not survey 10,000?

*Image placeholder: screenshot of the
poll report*

[img/ch03/poll_report.png](#)

And then the polls were wrong

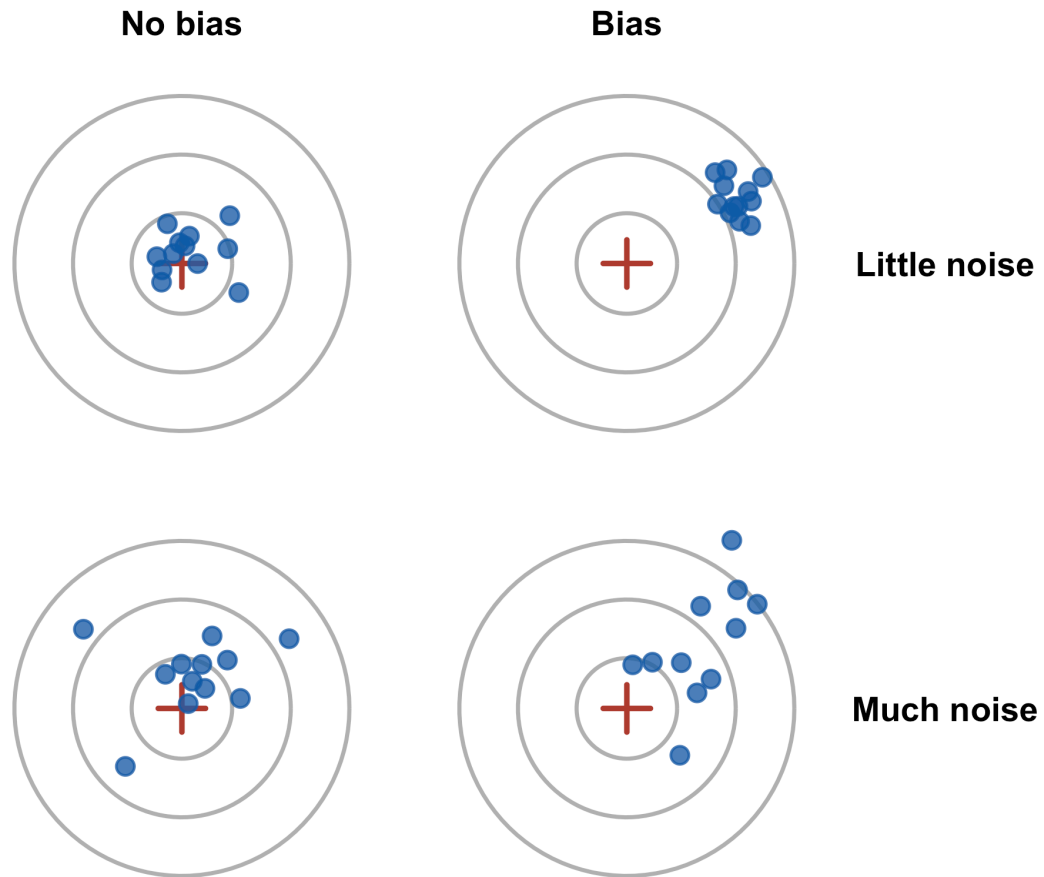
- 2016: state polls in several Midwestern states **underestimated Trump's support by more than their margins of error**
- Post-election analyses: late deciders; **voters without a college degree underrepresented** and not corrected for
- The margin of error did not warn — because it only covers **one kind of error**

*Image placeholder: headline or chart on the 2016
polling miss*

[img/ch03/polls_2016.png](#)

Kennedy et al. (2018), "An Evaluation of the 2016 Election Polls in the United States", *Public Opinion Quarterly*, 82(1), 1–33.

Two kinds of error



Random error (sampling variation)

- Each sample is a slightly different mix of people
- **Quantifiable** — and **shrinks** with larger samples
- The margin of error is about this kind

Systematic error (bias)

- The sample differs from the population *systematically*
- A larger sample **does not help** — “more confidently wrong” (\$1.6)

Why managers should care

- Satisfaction score up from 7.2 to 7.5?
- New landing page converts 3.4% instead of 3.1%?
- Ad A: CTR 1.9%, ad B: 2.2%?

The first question is not “*why?*” but “**how sure are we that anything changed at all?**” Every number from a sample is an **estimate** — and a different sample would give a different number.

3.2 Populations and samples

From the sample to the population

- **Population:** the whole group we want to learn about (all voters, all customers, all WU students)
- **Census:** data from every member — rare, expensive, often impossible
- **Sample:** a subgroup we actually observe
- **Statistical inference:** generalizing from the sample to the population

	Sample (statistic)	Population (parameter)
Size	n	N
Mean	$\bar{x}$	μ
Standard deviation	s	σ
Share	$\hat{p}$	p

Parameters: what we want to know. Statistics: what we can compute.

How the sample is drawn matters

	Probability sampling	Non-probability sampling
Principle	Known chance of selection; random	Convenience or researcher's judgment
Techniques	Simple random · stratified · cluster · systematic	Convenience · quota · judgmental · snowball
Uncertainty	Can be quantified	Cannot be quantified reliably

Typical sources of selection bias: *under-coverage* (not in the sampling frame) · *non-response* (non-respondents differ) · *voluntary response* (only the strongly opinionated answer)

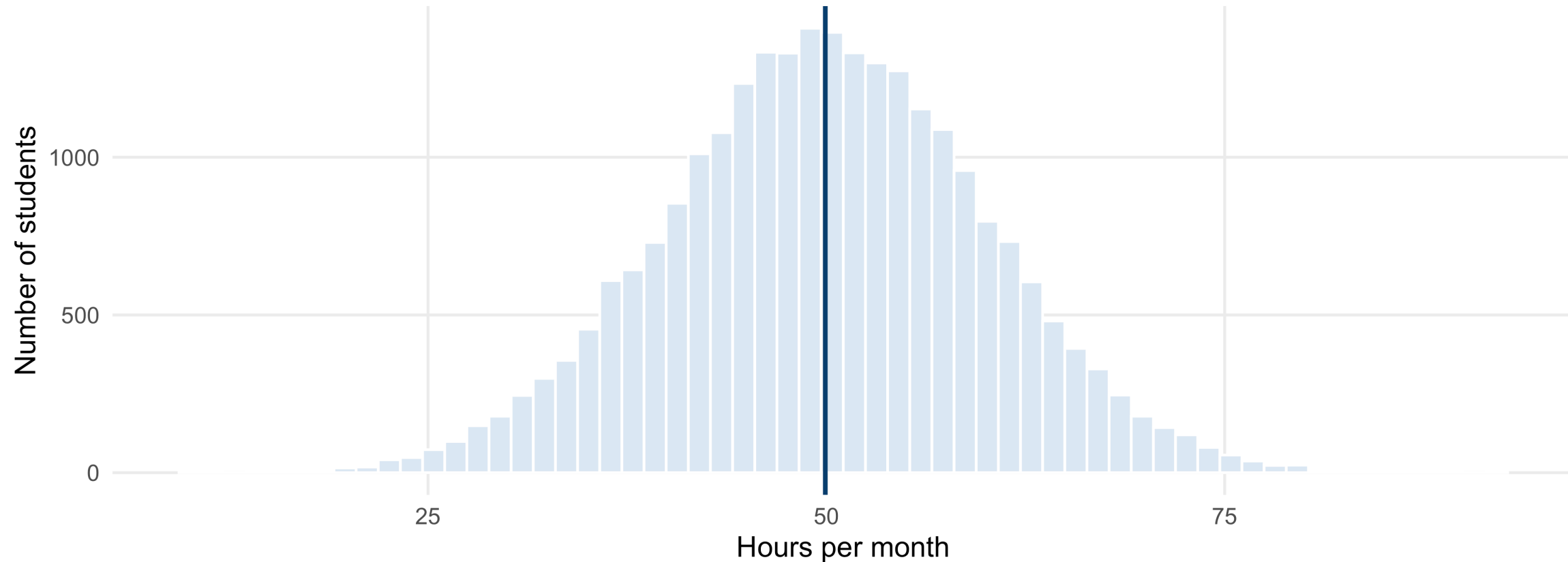
Your group project uses a convenience sample — fine for learning, but be honest about it. The good news: your **experiment** is still fair, because conditions are randomized *within* the sample (§1.6).

3.3 Sampling variation

A world in which we know the truth

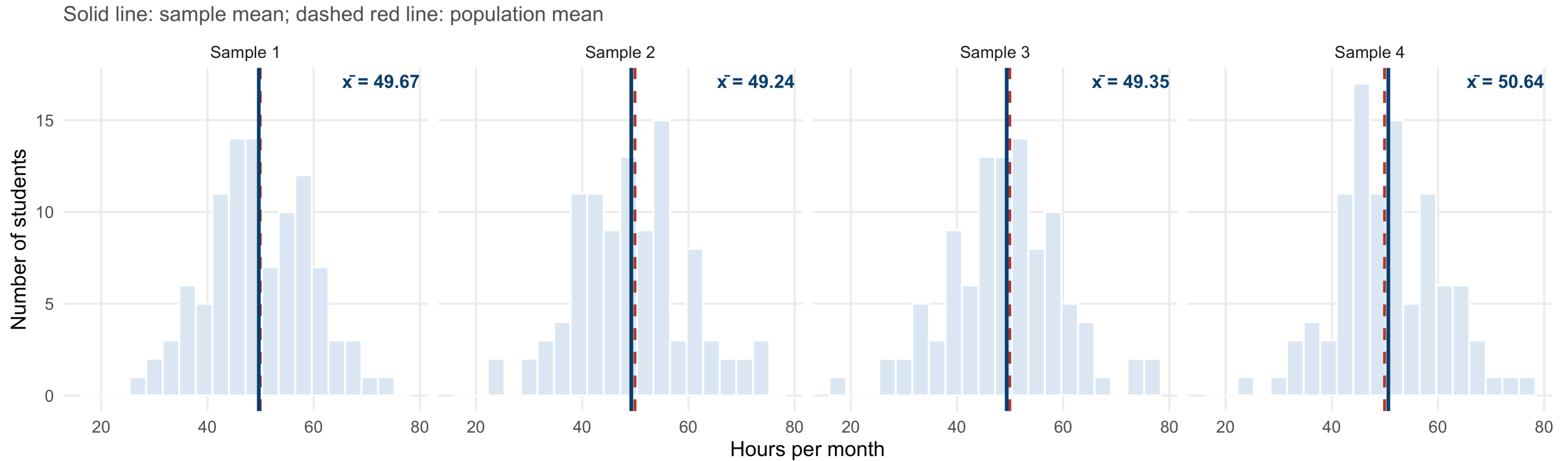
Monthly listening time of all 25,000 WU students

Population mean $\mu = 49.93$ hours; population SD $\sigma = 10.02$ hours



Monthly listening time of all 25,000 WU students — simulated, so we know the true values. In reality we never do: that is why we sample.

One sample, then another



Sampling variation: each random sample of 100 students gives a slightly different mean. Not a mistake — the unavoidable price of looking at a part instead of the whole. The question is only: **how large** is it?

Live demo: the sampling machine

Course book, §3.3 — draw 1, then 100, then 1,000 samples and watch the means pile up.

<https://wu-rds.github.io/MA2026/03-uncertainty.html#sec-unc-sampling>

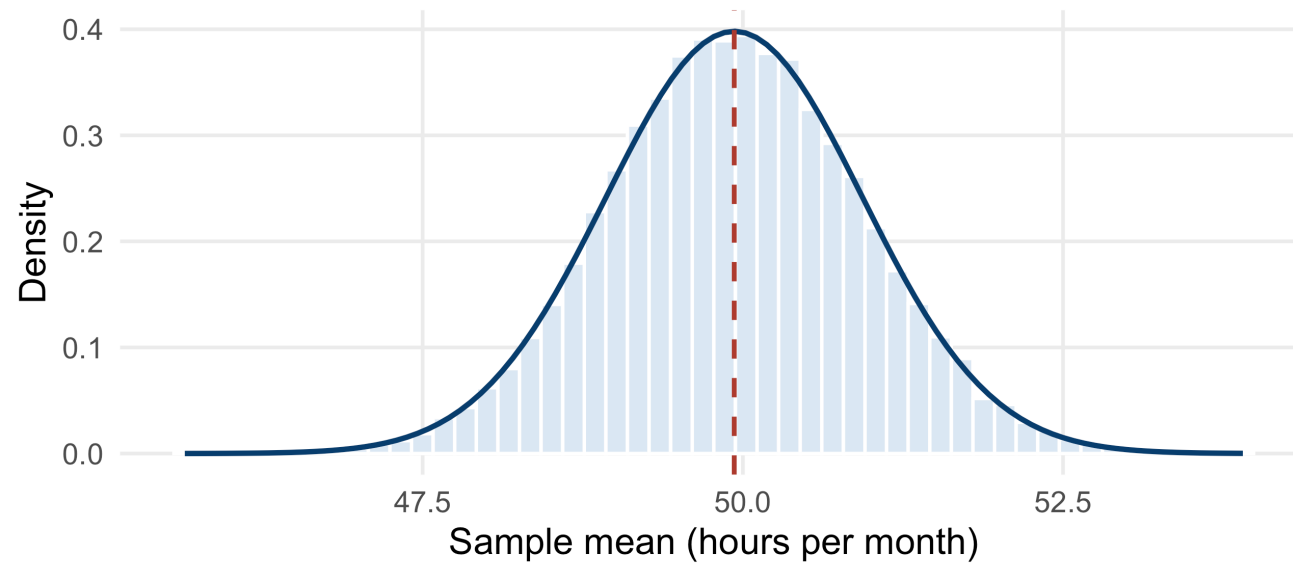
Screenshot placeholder: the sampling-machine widget
`img/ch03/widget_sampling.png`

Before clicking “1,000 samples”: where will the sample means pile up — and how far from the true mean will the most extreme ones be?

The sampling distribution

Means of 20,000 samples with $n = 100$ students each

Mean of the sample means = 49.93; SD of the sample means = 1.00



The means of *many* samples:

1. **Centered on the truth** — no over- or underestimation on average
2. **A known width** — the *standard error*
3. **A known shape** — normal (even for non-normal data, \$3.5)

We only ever have **one** sample — but these three properties tell us how much to trust it.

3.4 The standard error

How precise is a sample mean?

The width of the sampling distribution:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

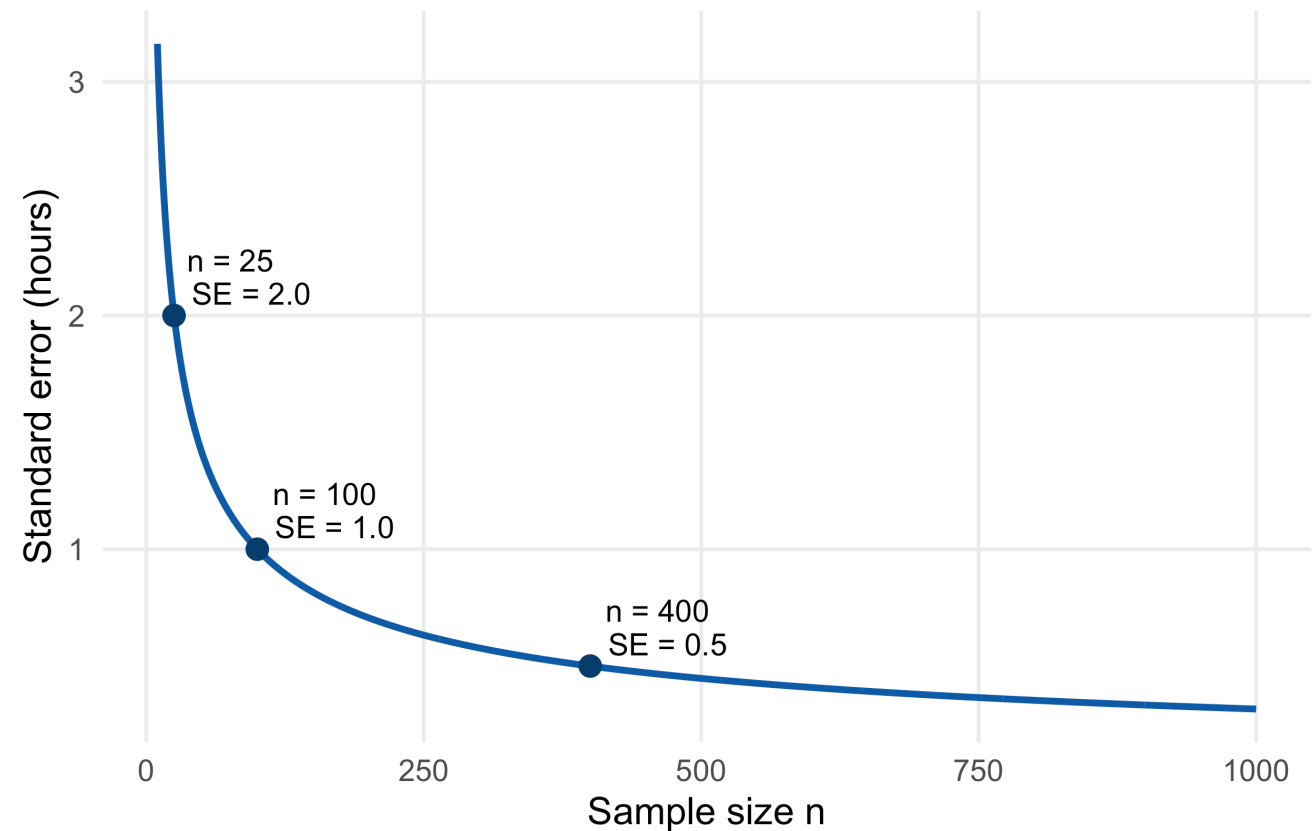
- σ : variability in the population
- n : sample size

Our example: $10 / \sqrt{100} = 1.00$ hours

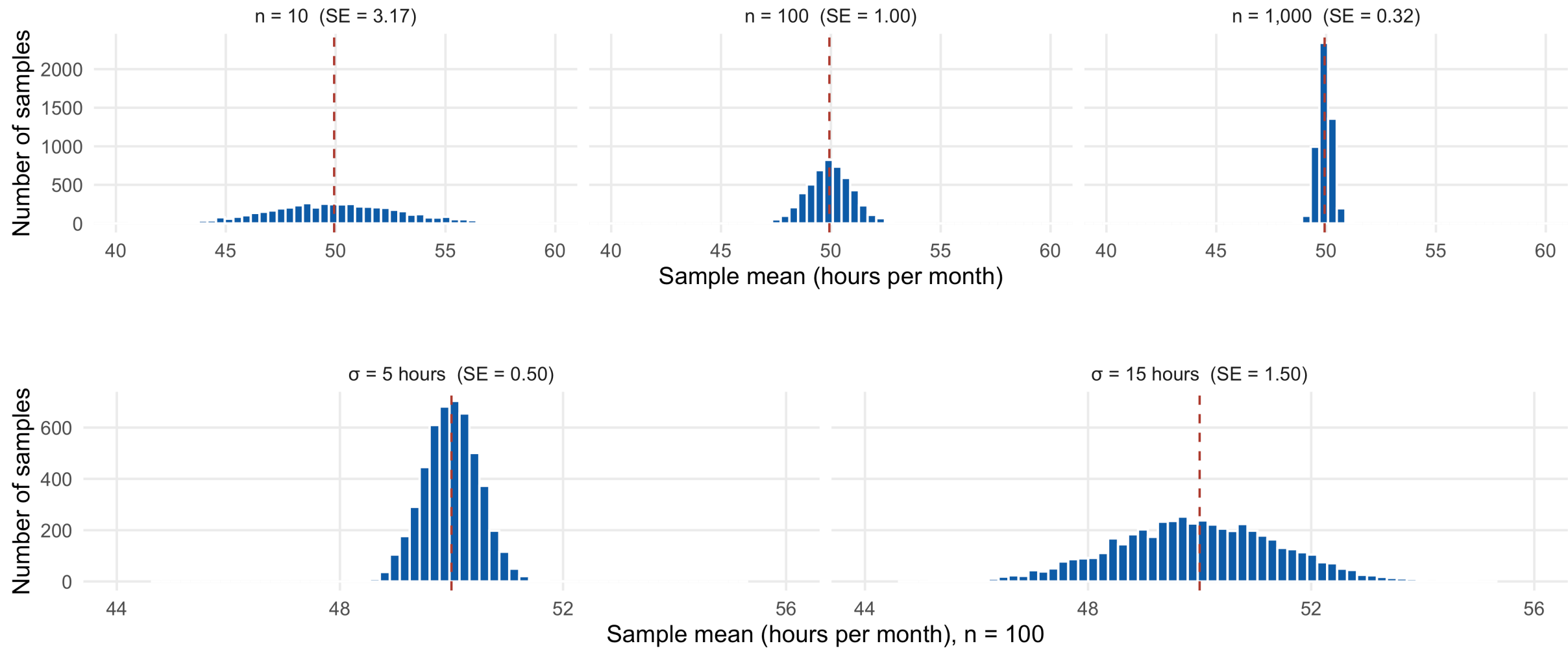
Square-root law: four times the sample, half the standard error — **diminishing returns**. This is why polls survey ~1,000 people, not 10,000.

Four times the sample size, half the standard error

Standard error of the mean for a population SD of $\sigma = 10$ hours



What drives the standard error



Larger samples → sample means cluster more tightly. **More variable populations** → sample means scatter more.

Live demo: what drives the standard error?

Course book, §3.4 — move the sliders for n and σ .

<https://wu-rds.github.io/MA2026/03-uncertainty.html#sec-unc-se>

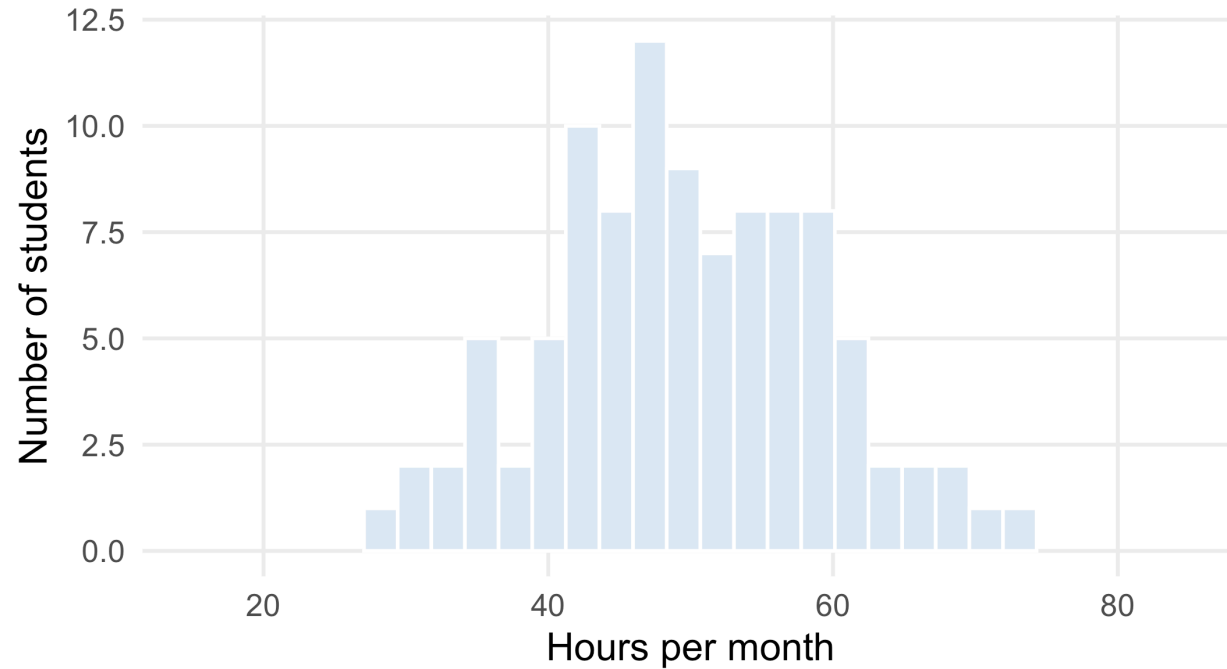
Screenshot placeholder: the standard-error widget
`img/ch03/widget_se.png`

Your survey has 100 respondents and a standard error of 2 points. Your boss wants **1** point. How many respondents do you need?

Standard deviation vs. standard error

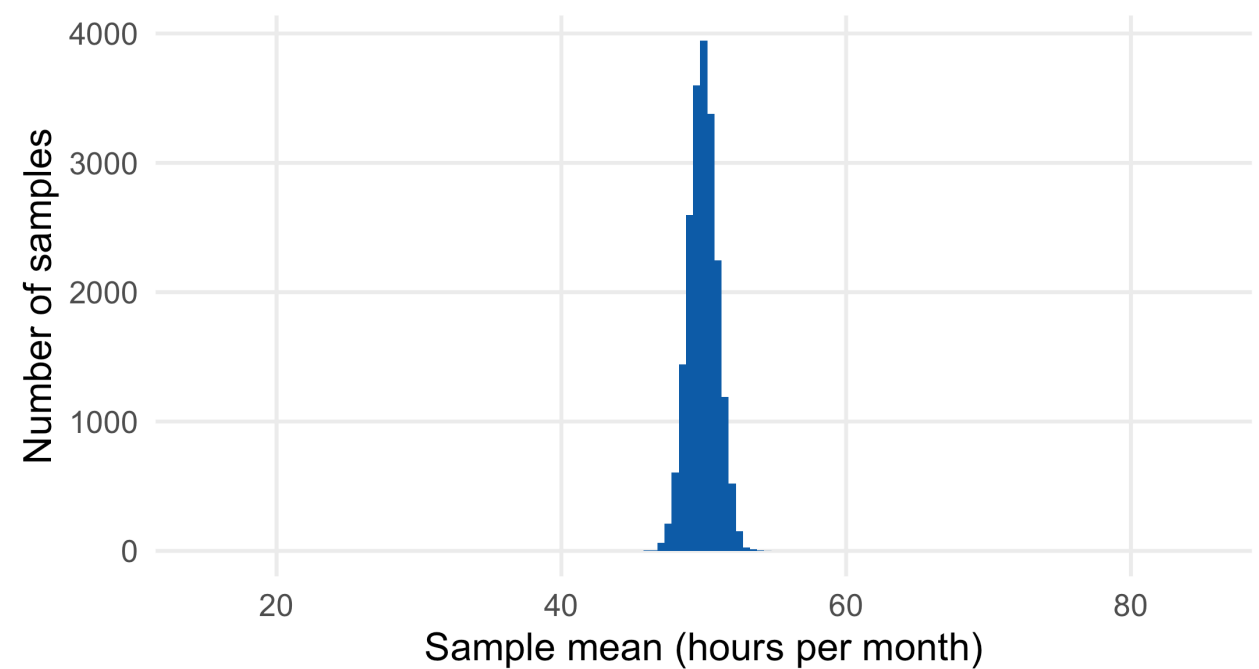
One sample: how students differ

Standard deviation $s = 9.4$ hours



Many samples: how sample means differ

Standard error = 1.0 hours



SD: how much **individuals** differ. A property of the data — does *not* shrink with more data.

SE: how much **sample means** would differ. The precision of the estimate — shrinks with $\sqrt{n}$.

In practice: estimate σ with s

We never know σ — so we use the sample standard deviation s :

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

Our first sample: $s = 10.44 \rightarrow SE = 1.04$ hours (true value: 1.00).

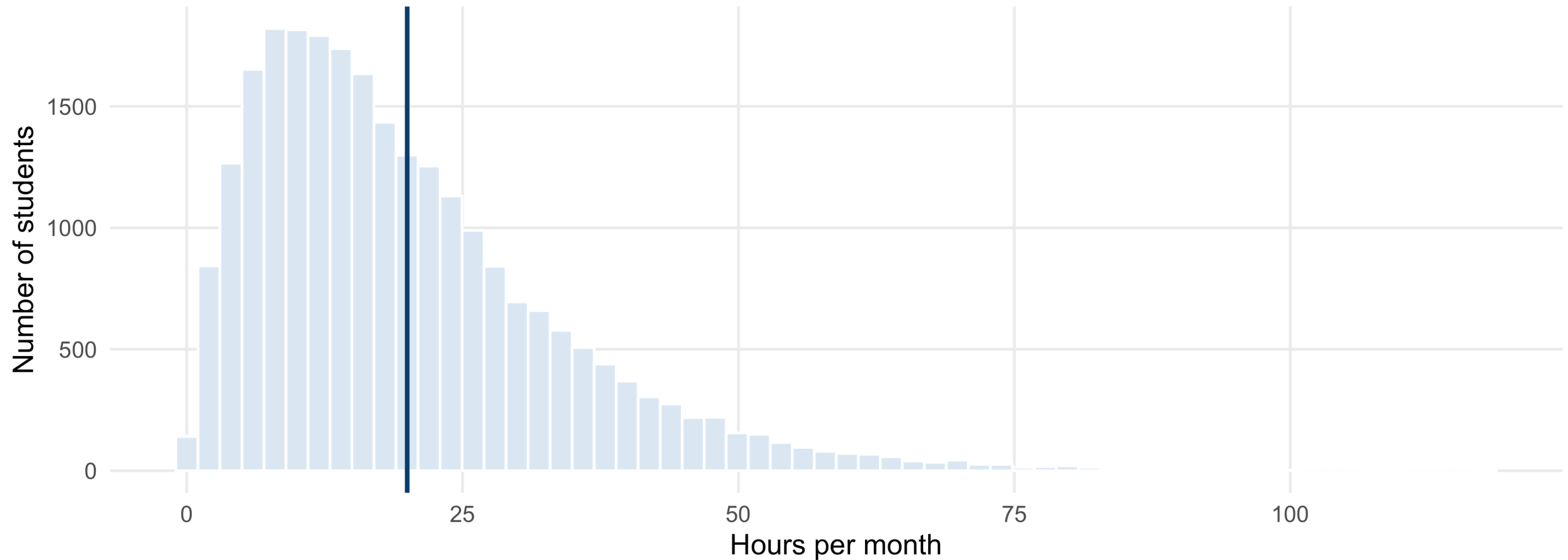
Which one to report? Describe how customers differ $\rightarrow$ **SD**. Describe how precisely you estimated the mean $\rightarrow$ **SE** or a **confidence interval**. Error bars in charts must say which one they show.

3.5 The central limit theorem

What if the data are not normal?

A right-skewed population of listening times

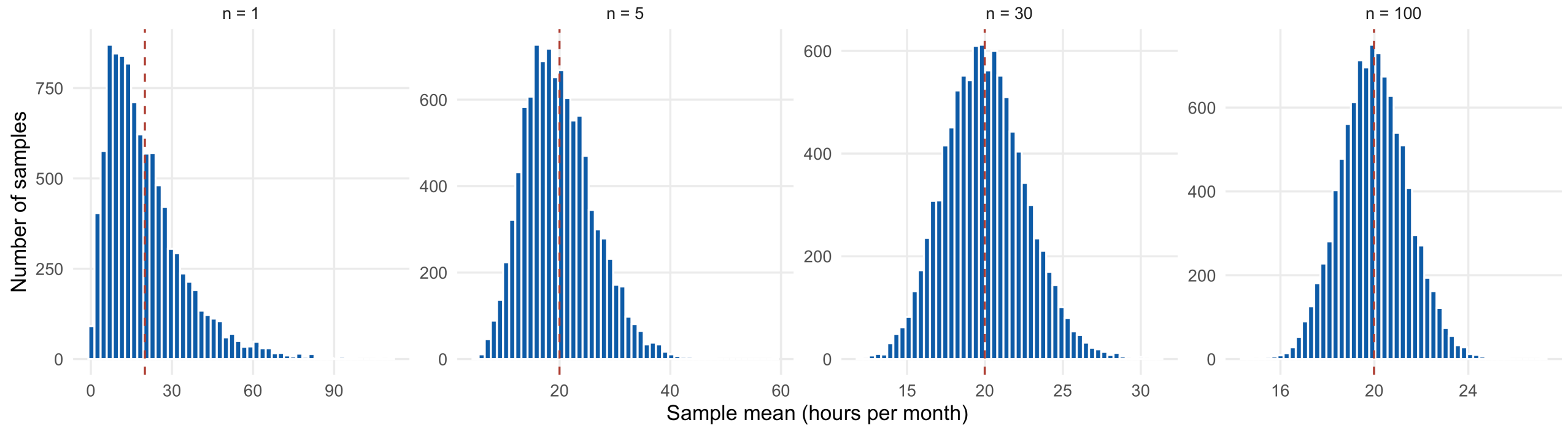
Population mean $\mu = 19.98$ hours; population SD $\sigma = 14.15$ hours



Marketing data are rarely normal: sales, visits, streams, revenue — **most customers a little, a few a lot.**

Sample means become normal anyway

Distribution of 10,000 sample means for different sample sizes n (skewed population)



Central limit theorem: whatever the shape of the population, the **means** of samples become normally distributed as n grows — centered on μ , with width $\sigma/\sqrt{n}$.

Live demo: the central limit theorem

Course book, §3.5 — choose a population shape, increase n step by step.

<https://wu-rds.github.io/MA2026/03-uncertainty.html#sec-unc-clt>

Screenshot placeholder: the CLT widget (two-group population)

`img/ch03/widget_clt.png`

Why it matters: we do not need normal *data* — we need a normal *sampling distribution*. “ $n \geq 30$ ” is a rule of thumb, not a law: for extremely skewed data (streams!) you need much more.

3.6 Confidence intervals

From a point to an interval

Point estimate: “the average listening time is 49.2 hours”

Interval estimate: “... between 47.2 and 51.3 hours”

Casting a net: we cannot see the true mean — but we can throw a net around our estimate that catches it most of the time.

$$\text{CI} = \text{estimate} \pm \underbrace{\text{critical value}}_{\text{how sure we want to be}} \times \underbrace{\text{standard error}}_{\text{how much the estimate wobbles}}$$

Net metaphor after Radziwill (2015), *Statistics (The Easier Way) with R*.

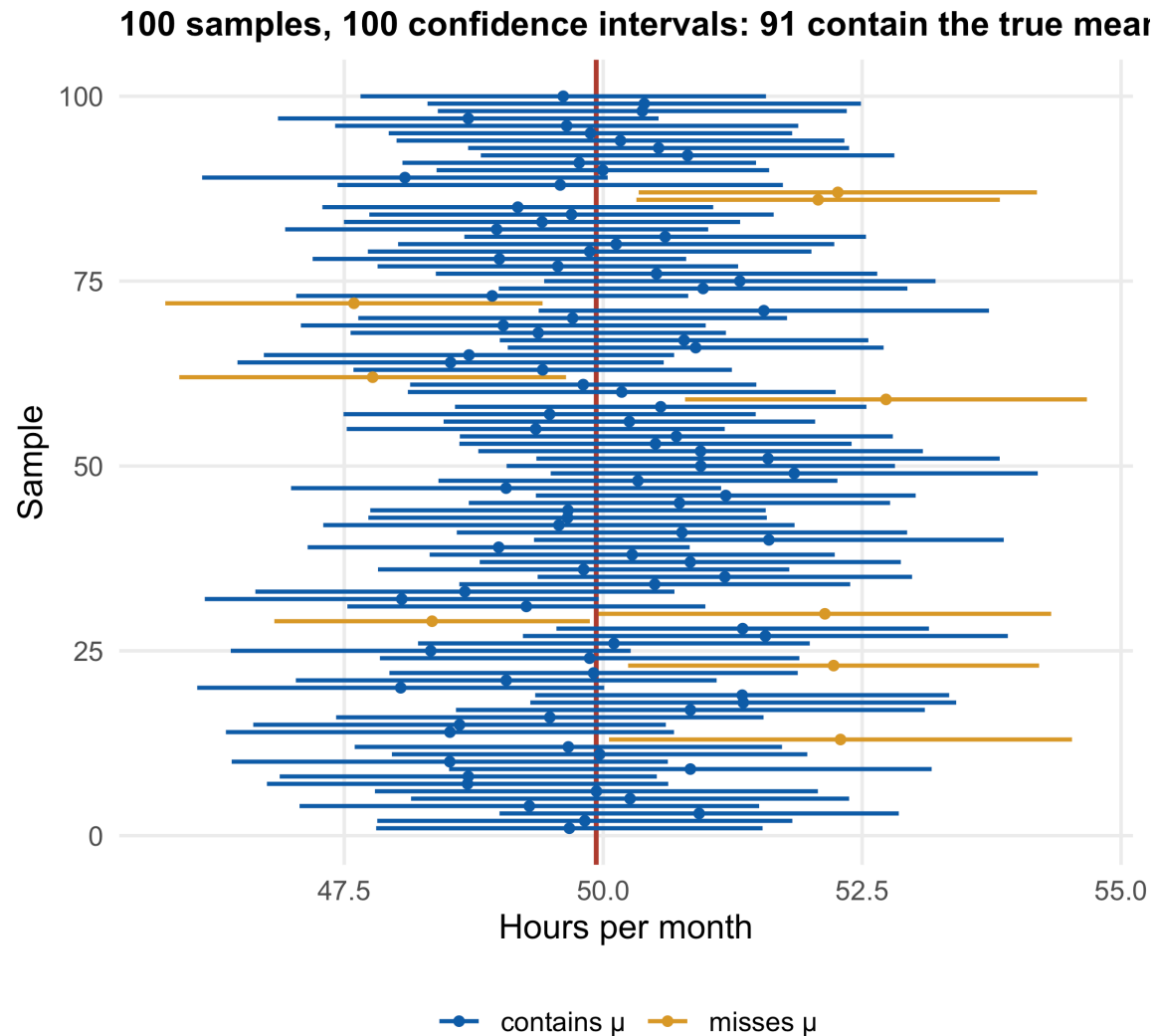
Constructing a 95% confidence interval

- Sample means are normal around μ with $SD = SE \rightarrow 95\%$ lie within $\pm 1.96 SE$
- Turned around: $\bar{x} \pm 1.96 \times SE$ catches μ in 95% of all samples

Our sample: $49.24 \pm 1.96 \times 1.04 = [47.19; 51.29]$ hours

True mean: 49.93 — inside. This time.

What “95% confidence” means



The confidence refers to the **method**: of all intervals constructed this way, **~95% contain** the true mean.

Not quite right: “95% probability that μ is in *this* interval” — a specific interval either contains μ or not.
Fine in practice: “we are 95% confident that ...” — remembering that ~1 in 20 such intervals is wrong.

Live demo: how often does the net catch the truth?

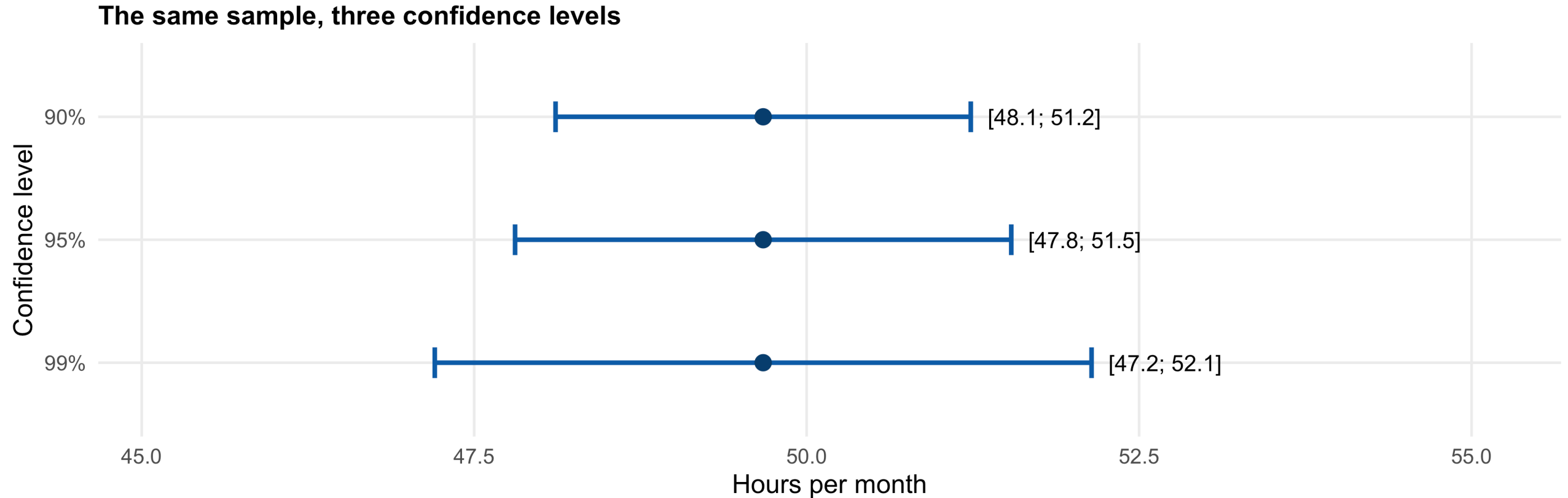
Course book, §3.6 — draw batches of 100 intervals; change n and the **confidence level**.

<https://wu-rds.github.io/MA2026/03-uncertainty.html#sec-unc-ci>

Screenshot placeholder: the coverage widget
`img/ch03/widget_ci.png`

What happens to the **width** and to the **number of misses** when you (a) increase n , (b) switch from 95% to 99%?

Confidence level vs. width



- More confidence → **wider** net (99%: 2.58 instead of 1.96)
- Narrow **and** reliable → only via **larger n**

<i>n</i>	10	30	100	∞
<i>t</i> value (95%)	2.26	2.05	1.98	1.96

Small samples: *t*-distribution — R does it automatically

3.7 Shares and the margin of error

Decoding “± 3.1 percentage points”

Standard error of a share:

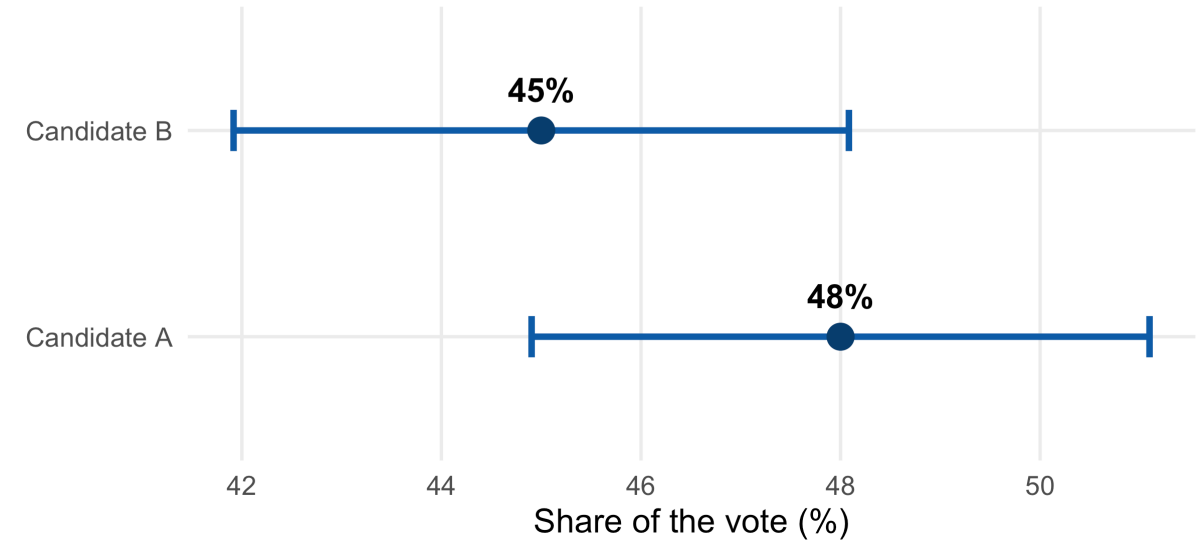
$$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Margin of error = $1.96 \times SE_{\hat{p}}$, largest at $\hat{p} \approx 50\%$:

$$1.96 \times \sqrt{\frac{0.5 \times 0.5}{1,000}} = \mathbf{0.031}$$

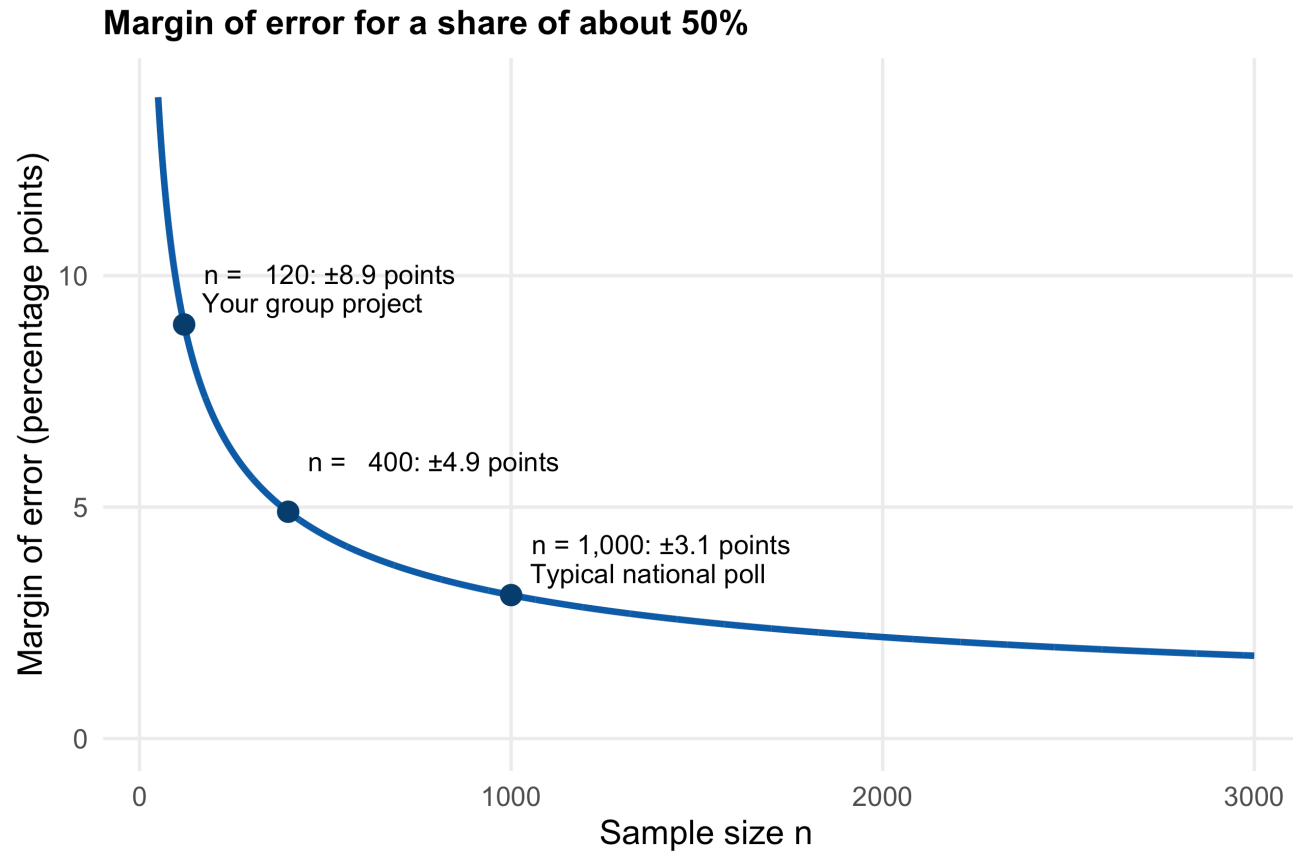
A poll of 1,000 voters: 95% confidence intervals

The intervals overlap — but that alone does not settle whether the lea



The margin of error of the **lead** is roughly **twice** as large: a 3-point lead in this poll is far from safe.

How many respondents do you need?



Your group project (~120 respondents): a share of ~50% comes with **± 9 points**. Think twice before interpreting a 5-point difference.

Live calculator: course book, §3.7

<https://wu-rds.github.io/MA2026/03-uncertainty.html#sec-unc-proportions>

3.8 Uncertainty in R

A confidence interval for a mean

```
1 set.seed(6789)
2 population <- rgamma(25000, shape = 2, scale = 10) # the "unknown" population
3 listening <- tibble(hours = sample(population, size = 100))
```

```
1 n <- nrow(listening)
2 xbar <- mean(listening$hours)
3 se <- sd(listening$hours) / sqrt(n)
4 crit <- qt(0.975, df = n - 1)
5 c(mean = xbar, lower = xbar - crit * se, upper = xbar + crit * se)
```

```
      mean    lower    upper
19.77946 16.82950 22.72943
```

```
1 t.test(listening$hours)$conf.int
```

```
[1] 16.82950 22.72943
attr(,"conf.level")
[1] 0.95
```

`qt(0.975, ...)`: 2.5% in each tail. `t.test()` does it all in one line.

Shares and sample size planning

Share of students listening > 30 hours:

```
1 p_hat <- mean(listening$hours > 30)
2 se_p  <- sqrt(p_hat * (1 - p_hat) / n)
3 c(share = p_hat,
4   lower = p_hat - 1.96 * se_p,
5   upper = p_hat + 1.96 * se_p)
```

share	lower	upper
0.2200000	0.1388077	0.3011923

Respondents needed for ± 5 points:

```
1 me_target <- 0.05
2 ceiling(qnorm(0.975)^2 * 0.25 / me_target^2)
```

[1] 385

`prop.test(x, n)$conf.int` gives a refined interval for shares directly.

What confidence intervals do not tell you

Three limits

1. **Random error only.** A biased sample gives a precise but wrong estimate — a larger biased sample, a more precise and equally wrong one (2016 polls; \$1.6; \$2.5)
2. **Precision, not spread.** A narrow interval for the *average* does not mean customers are alike
3. **A statement about a procedure.** ~1 in 20 intervals misses — compute many, and some will

Next: an A/B test shows a difference between two groups. Is it larger than sampling variation alone would produce? The difference has its own standard error — **the logic of today carries over one to one.**

Learning check

1. A poll of 1,000 voters reports ± 3.1 points. How is this number calculated? Name one error it does **not** cover.
2. You double your survey from 400 to 800 respondents. By what factor does the standard error change?
3. Random sample of 180 users: mean listening time 7.34 hours/week, SD 6.87. Compute the 95% CI (critical value 1.96).
4. Your website survey (anyone can answer) has 20,000 responses and a very narrow CI. Why might it still mislead?